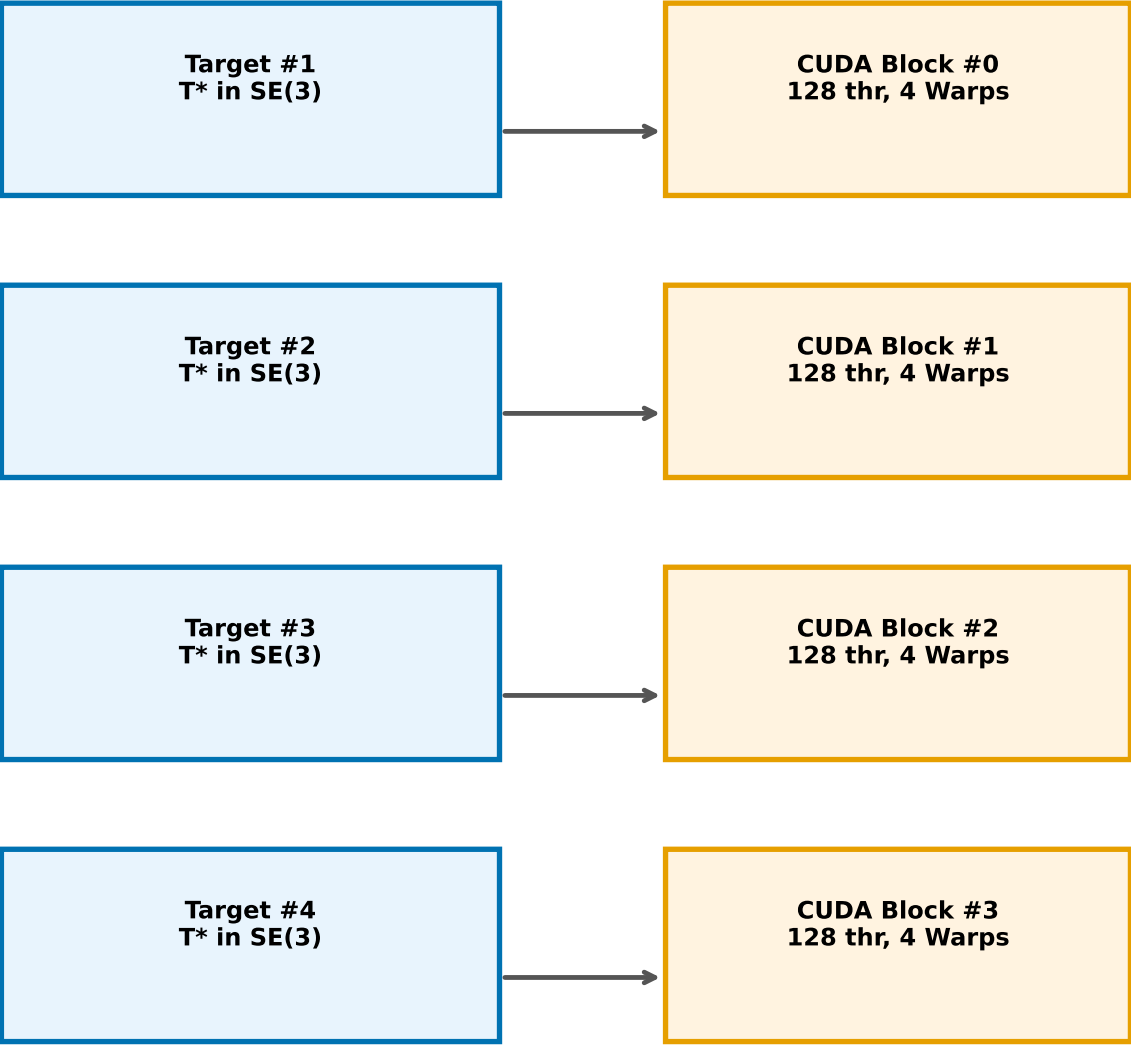


CUDA Small-Matrix Acceleration Architecture: 1-Block-per-Target + Register LDLT + PaddedMat6x8



Single Block Internal: 128-Thread 4-Warp Pipeline for 6x6 Small-Matrix Acceleration

[Storage Hierarchy]

- Constant Memory: UR10 model params (broadcast, zero-latency)
- Shared Memory: PaddedMat6x8 (stride=8, zero bank conflict)
- Registers: LDLT 6x6 solver (~60 regs, ~0.1us, zero mem access)

[Compute Pipeline per DLS Iteration]

- FK: Lane 0 (Rodrigues, constant-mem reads)
- Jacobian 6 cols: Lane 0-5 (6-way parallel, central diff)
- Hessian 36 elems: Lane 0-35 (36-way parallel, weighted inner prod)
- Gradient 6 dims: Lane 0-5 (6-way parallel)
- LDLT Solve: Lane 0 (register-resident, ~63 scalar ops)
- Joint Update: Lane 0-5 (clamp + branch align)

Output

- * Best q (6 joints)
- * Pose Error (pos+rot)
- * Iteration Count
- * Limit Status
- * NaN/Inf = 0

NVIDIA GeForce RTX 4060 Laptop GPU (Ada Lovelace SM89) | CUDA 13.3 | 128 threads/block

Key innovations: PaddedMat6x8 (bank-conflict-free shared mem) | Register LDLT (0.1us) | FP32/FP64 Mixed Precision (2.5x speedup) | 1-Block-per-Target Mapping